Measurable, Self-Improvable, and Perceptually Capable: Building the Full Stack of Intelligent AI Systems

The rapid rise of foundation models has created an urgent need to answer three fundamental questions that determine whether AI systems are truly useful, reliable, and capable of continued progress: *How do we rigorously measure what AI systems can do? How do we systematically improve AI capabilities beyond the ceiling of human-annotated supervision?* And *how do we build AI systems that perceive and act intelligently across the diverse multimodal environments of the real world?*

My research program addresses these three questions in concert, spanning **scalable evaluation**, **reinforcement learning with verifiable rewards**, and **multimodal agentic systems**. Critically, these are not independent tracks — they form a **mutually reinforcing cycle**: rigorous evaluation makes scientific progress legible and reproducible; automated evaluation signals, when scaled, become the reward signals that power reinforcement learning beyond supervised ceilings; and as RL enables increasingly capable systems, those systems must operate as autonomous agents that perceive and act across richly multimodal, real-world environments. My work across these three pillars has accumulated **3,000+ citations**, with publications at ACL, EMNLP, CVPR, NeurIPS, ICLR, and TMLR.

This research cycle began with the observation that evaluating LLM outputs at scale requires fundamentally new methodology. Standard metrics are scalar, opaque, and poorly correlated with human judgment. I developed ensemble-based pairwise selection for combining the complementary strengths of diverse LLMs [1] and explainable, reference-free evaluation metrics that provide natural-language error analysis alongside scores for any text generation task [2]. As AI rapidly expanded into visual domains, I extended these principles to conditional image synthesis [3] and video generation [4, 5], and deployed open community evaluation platforms where real users signal model preferences at scale [6, 7]. Recognizing that evaluation must stress-test the full breadth of model capabilities, I contributed to landmark multimodal benchmarks that challenge models with college-level expert knowledge [8] and developed comprehensive evaluation suites spanning hundreds of real-world tasks and diverse output formats — from free-form text to JSON, code, LaTeX, and SVG [9, 10].

With reliable evaluation in place, a natural insight emerges: **automated evaluation signals can serve directly as reward signals for reinforcement learning**. Supervised fine-tuning is fundamentally bounded by the availability and quality of human-labeled data. My RL research establishes that scalable synthesis of verifiable signals — automated test cases for code [11], web-crawled domain questions with generative verification [12], and critique-based feedback [13] — can push model capabilities far beyond supervised ceilings. These training innovations culminate in a unified, modular framework for agentic reinforcement learning with diverse tool use [14], enabling LLMs to execute multi-step trajectories across code execution, web search, SQL databases, and visual perception. At the frontier, an open 30B agentic model trained on synthesized long-horizon research trajectories demonstrates that these techniques together can surpass proprietary systems including GPT-4.1, Claude Opus 4, and Gemini-2.5-Pro on deep research tasks [15].

The third pillar of my research recognizes that agents operating in the real world do not perceive text alone — they must reason across interleaved images, long videos, documents, user interfaces, and diverse file systems. I established that strong multi-image reasoning can be acquired through targeted instruction tuning at academic scale, without massive web pretraining [16], and that system-algorithm co-design can make long-video understanding practical in real-time settings on commodity hardware [17]. Looking ahead, I aim to extend this multimodal foundation toward agents that operate across the full spectrum of real-world digital environments — from structured document processing and computer use to heterogeneous sensor inputs.

Together, these three pillars constitute a coherent agenda for building AI systems that are **measurable**,

self-improvable, and **perceptually capable** — the three properties I believe are foundational to the next generation of intelligent systems. My research contributions are structured across three areas:

§1: Scalable and Interpretable Evaluation — ***Establishing reliable metrics and benchmarks for LLMs, multimodal models, and generative systems***

§2: Reinforcement Learning and Agentic Systems — ***Advancing LLM capabilities through reward design, training paradigm innovation, and agentic RL infrastructure***

§3: Multimodal Foundation Models and Multimodal Agents — ***Building capable multimodal architectures and extending them toward real-world agentic settings***

§1 Scalable and Interpretable Evaluation of AI Systems

A prerequisite for systematic improvement is systematic measurement. My evaluation research moves from ranking and selecting among model outputs, to explaining what goes wrong, to deploying evaluation at community scale, to building comprehensive capability benchmarks — forming a complete measurement stack for the AI development cycle.

From Scalar Scores to Pairwise Ranking and Ensembling. The observation that different LLMs excel on different inputs motivates a fundamentally different evaluation paradigm: pairwise comparison rather than independent scoring. I developed *LLM-Blender* [1], an ensembling framework comprising *PairRanker* — a cross-attention encoder that jointly encodes input–candidate pairs to determine fine-grained preferences — and *GenFuser*, which synthesizes top-ranked outputs into improved results. LLM-Blender significantly outperforms individual models and establishes pairwise comparison as a scalable, reliable paradigm for both evaluation and output selection, with over 640 citations and integration into subsequent preference-learning pipelines.

Explainable, Instruction-Guided Metrics. Pairwise comparison tells us which output is better but not *why*. I developed *TIGERScore* [2], the first instruction-guided, reference-free evaluation metric that generates natural-language error analysis alongside scores for arbitrary text generation tasks. Trained on *MetricInstruct* — 42K quadruples of (instruction, input, output, error analysis) spanning 6 tasks and 23 datasets — TIGERScore achieves open-source state-of-the-art correlation with human ratings and even surpasses reference-based metrics, with human judges finding its explanations 70.8% accurate. TIGERScore demonstrates that evaluation can be simultaneously general, explainable, and reference-free.

Extending Evaluation to Multimodal Generative Systems. As generative AI expanded to images and video, reliable evaluation became an acute challenge — FID, CLIP, and FVD fail to capture nuanced human preferences. I developed *VIEScore* [3] for conditional image synthesis, leveraging multimodal LLMs as zero-shot judges across seven generation tasks, achieving human-level correlation (0.4 Spearman vs. 0.45 human–human). For video, I built *VideoScore* [4] on *VideoFeedback*, the first large-scale human-annotated multi-aspect scoring dataset across 37.6K synthesized videos from 11 generative models, achieving 77.1 Spearman correlation — outperforming prior metrics by ∼50 points. *VideoScore2* [5] further introduces chain-of-thought reasoning before scoring, substantially improving reliability on complex video quality dimensions.

Community-Scale Evaluation Platforms. Benchmarks reflect the perspectives of their creators; reliable evaluation requires aggregating diverse human preferences at scale. I co-developed *GenAI Arena* [6], an open platform accumulating 6,000+ community votes across 27 open-source image, video, and editing models, revealing that even GPT-4o achieves only a 0.22 Pearson correlation when mimicking human visual-quality judgment. I co-developed *WildVision* [7], collecting real-world human preferences over VLM outputs in an open arena and producing *WV-Bench* — achieving a 0.94 Spearman correlation with arena Elo — surfacing systematic failure modes in hallucination, spatial reasoning, and expert-domain knowledge that curated benchmarks miss.

Comprehensive Capability Benchmarks. I contributed to *MMMU* [8] (CVPR 2024 **Best Paper Finalist & Oral**, 1,935 citations), a landmark benchmark of 11.5K multimodal questions across 30 subjects at college-level difficulty, demonstrating that frontier models including GPT-4V and Gemini Ultra achieve only 56–59% accuracy. Addressing breadth, I co-developed *MEGA-Bench* [9] (ICLR 2025), scaling multimodal evaluation to 505 real-world tasks, diverse output formats (numbers, code, JSON, coordinates, LaTeX), and 40+ task-specific metrics with fine-grained capability reports. For structured outputs critical in software integration, I co-developed *StructEval* [10] (TMLR 2026), covering 18 formats and 44 task types, revealing that even o1-mini achieves only 75.58 average score and open-source models lag ∼10 points behind.

Future: Evaluating Agents in the New Era. Current benchmarks evaluate models on static, independent queries. The next frontier is evaluating autonomous agents across dynamic, multi-step, tool-augmented, and multimodal interaction settings. I plan to develop evaluation frameworks specifically designed for agents — assessing planning coherence, tool-use correctness, multi-step reasoning fidelity, and real-world task completion. This includes live evaluation platforms (extending GenAI Arena and WildVision) that collect human preferences over agent *trajectories*, and principled metrics that assess the quality of entire decision sequences rather than isolated outputs.

§2 Reinforcement Learning and Agentic Systems

Supervised fine-tuning is bounded by human annotation bottlenecks and cannot exceed the quality of its training labels. My second research thrust establishes a paradigm shift: **scalable synthesis of verifiable reward signals unlocks reinforcement learning that systematically improves AI capabilities beyond supervised ceilings**, and a modular agentic RL infrastructure enables this paradigm to scale to complex, multi-step, tool-augmented tasks.

Reward Design through Automated Data Synthesis. The central barrier to RL for LLMs is reliable reward. I developed *ACECODER* [11] (ACL 2025), which addresses this challenge in the code domain by constructing a large-scale pipeline that synthesizes (question, test-case) pairs from existing code corpora. Test-case pass rates provide an objective, scalable reward for training reward models via Bradley-Terry loss, achieving 10-point improvements for Llama-3.1-8B and 5-point improvements for Qwen2.5-Coder-7B through best-of-32 sampling, bringing 7B models to parity with 236B DeepSeek-V2.5. Full RL training from base models improves HumanEval+ by over 25% in just 80 optimization steps. ACECODER establishes test-case synthesis as a scalable, reliable reward paradigm, catalyzing a growing line of work in automated reward synthesis.

Extending this paradigm beyond code, I co-developed *General-Reasoner* [12] (NeurIPS 2025), which tackles the dual challenge of data scarcity and answer format diversity across physics, chemistry, finance, and law. General-Reasoner introduces: (1) a large-scale, web-crawled dataset of domain questions with verifiable answers, and (2) a generative model-based answer verifier replacing brittle rule-based checking with chain-of-thought, context-aware judgment. Evaluated across 12 benchmarks spanning MMLU-Pro, GPQA, SuperGPQA, TheoremQA, BBEH, and AMC, General-Reasoner demonstrates that the RL reasoning paradigm generalizes across domains when paired with scalable data collection and flexible verification.

Training Paradigm Innovation: Critique Reinforcement Learning. Standard RL trains models to generate correct outputs but does not develop the capacity for self-correction — a key component of human expertise. I co-developed *Critique-Coder* [13] (ICLR 2026), which introduces *Critique Reinforcement Learning* (CRL): the model generates critiques of (question, solution) pairs, rewarded solely by whether its final judgment matches the ground truth. By mixing 20% CRL data into standard RL training, Critique-Coder consistently outperforms RL-only baselines. Critique-Coder-8B achieves over 60% on LiveCodeBench v5, surpassing DeepCoder-14B and GPT-o1, while also demonstrating improved general reasoning on BBEH, confirming that critique ability transfers beyond code.

Unified Agentic RL Framework. Scaling RL to agentic settings — where models must invoke external tools, manage multi-turn interactions, and operate across heterogeneous observation modalities — requires principled infrastructure. I developed *VerlTool* [14] (ICLR 2026, 872⋆ GitHub), a unified, modular framework for agentic reinforcement learning with tool use. VerlTool formalizes multi-turn agentic trajectories with multi-modal observation tokens (text, image, video) and contributes four key design principles: upstream alignment with VeRL for simplified maintenance; standardized APIs supporting code execution, web search, SQL, and vision tools; asynchronous rollout execution achieving near 2× speedup by eliminating synchronization bottlenecks; and comprehensive evaluation across 6 agentic domains. VerlTool provides a scalable foundation for the broader research community to experiment with tool-augmented RL without domain-specific engineering overhead.

Long-Horizon Deep Research Agents. The frontier of agentic capability lies in long-horizon tasks requiring hundreds of multi-step, tool-augmented interactions with real-world information sources. I co-developed *OpenResearcher* [15] (2026), a fully open 30B-A3B agentic LLM trained on 96K high-quality deep research trajectories synthesized using native browser tools. OpenResearcher achieves 54.8% on BrowseComp-Plus, surpassing GPT-4.1, Claude Opus 4, Gemini-2.5-Pro, DeepSeek-R1, and Tongyi-DeepResearch, while being fully open-source in model weights, training data, and training pipeline. OpenResearcher demonstrates the viability of high-quality trajectory synthesis as a data strategy for complex agentic capabilities.

Future: Scaling Agentic RL and Multi-Agent Training. My future research will advance agentic RL along three axes. First, I will develop more sophisticated reward design for long-horizon tasks, including process-level rewards that assess intermediate reasoning steps rather than only final outcomes. Second, I will investigate multi-agent RL settings — such as a critic agent supervising a generator, a planner coordinating executors, or adversarial red-teaming — extending the self-improvement paradigm beyond single-agent RL. Third, I will develop curriculum and continual learning strategies that enable agents to tackle progressively harder, open-ended tasks, moving toward systems that improve autonomously over time.

§3 Multimodal Foundation Models and Multimodal Agents

Real-world AI agents do not operate on text alone. The next generation must perceive and reason across diverse modalities — images, video, documents, user interfaces, file systems, and beyond — and integrate multimodal perception with agentic action. My third research thrust builds the multimodal foundation for this vision.

Efficient Multi-Image Reasoning through Instruction Tuning. Multi-image understanding — co-referencing, comparing, and reasoning across sets of images — is fundamental to real-world vision tasks yet had been assumed to require massive web-scale pretraining. I developed *MANTIS* [16] (**TMLR 2025 Outstanding Paper Award, 1/1,539 selected**), which establishes that targeted instruction tuning at academic scale is sufficient. I constructed *Mantis-Instruct*, 721K multi-image instruction examples spanning co-reference, comparison, temporal, and compositional reasoning. Mantis-SigLIP achieves state-of-the-art results on all five multi-image benchmarks, outperforming Idefics2-8B by 11 absolute points despite Idefics2 being pretrained on 200× more multi-image data. MANTIS demonstrates that high-quality, targeted data is more important than data volume — a principle that generalizes to data synthesis strategies for multimodal agents.

Real-Time Long Video Understanding. Long video presents dual bottlenecks: sequential decoding that can take over a minute for hour-long inputs, and costly LLM prefilling of millions of tokens. I co-developed *QuickVideo* [17] (TMLR 2025), a system-algorithm co-design addressing both constraints. *QuickCodec* parallelizes CPU-based video decoding across keyframe-aligned intervals for 2–3× speedup; *QuickPrefill* applies KV-cache pruning for memory-efficient processing; an overlapping scheme pipelines CPU decoding with GPU inference. Together, QuickVideo reduces end-to-end latency by minutes, making real-time long-video understanding practical on commodity hardware — a prerequisite for video-perceiving agents in production settings.

Future: Multimodal Agents Across Diverse Real-World Environments. The multimodal perception capabilities established in MANTIS and QuickVideo address only a subset of modalities agents encounter in practice. Real-world digital environments present heterogeneous inputs: structured documents (PDFs, spreadsheets, code files), web pages and user interfaces, database outputs, and multi-sensor streams. My future research will build multimodal agents capable of perceiving and acting across this full diversity. This includes: developing training methodologies for document-grounded question answering and information extraction; building computer-use agents that perceive rendered UI screenshots and execute keyboard/mouse actions; and investigating how diverse modality representations can be unified into a shared agentic context window that supports tool-augmented reasoning. The RL and data synthesis infrastructure from §2 will provide the training backbone, while the evaluation frameworks from §1 will supply the benchmarks that make progress measurable — completing the cycle.

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